



TOKAR
SCIENTIFIC AND PRODUCTION ENTERPRISE

27 years on the market **9** patents **4** know-how

**HIGHLY EFFECTIVE
MULTIFUNCTIONAL
ADMIXTURE
TO CONCRETES**

D5 ЭКО

**FOR WATERPROOF CONCRETES
WITH INCREASED STRENGTH, SULFATE-RESISTANCE AND DURABILITY**



www.d5-tokar.ru

**We have been developing and producing
innovative admixtures to concretes
for more than 27 years**

1998 Start of production of IR-1 – the admixture for waterproof concretes

2000 Start of production of EXTRA – the admixture to increase the strength of concretes

2006 Start of production of the admixture D5 for waterproof concretes and the admixture D11 for high strength concretes

2012 Start of production of the improved admixture **D5** which combines properties of the admixtures D5 and D11

2017 Start of export deliveries of D5 to [South Korea](#), [Kazakhstan](#) and [New Zealand](#)

2021 Start of export deliveries of D5 to Azerbaijan

2022 Start of export deliveries of D5 to Belarus

2023 Start of export deliveries of D5 to Saudi Arabia and Vietnam

2024 Start of export deliveries of D5 to Kyrgyzstan

2025 Start of export deliveries of D5 to Abkhazia



!

- **No cement**
- **No microsilica**
- **No fly ash**
- **No chlorine**



What properties does D5 impart to concretes?



**HIGH
WATERPROOFING**



**HIGH
SULFATE-RESISTANCE**



**ACCELERATION OF
STRENGTH GAIN**



**INCREASE OF
STRENGTH**



**INCREASE OF
FROST-RESISTANCE**



**PLASTICITY
COHESION
PUMPABILITY**



AIR ENTRAINMENT



**SELF-SEALING OF
THROUGH CRACKS
IMPERMEABILITY OF
COLD JOINTS**



ECO-FRIENDLINESS

W34 !

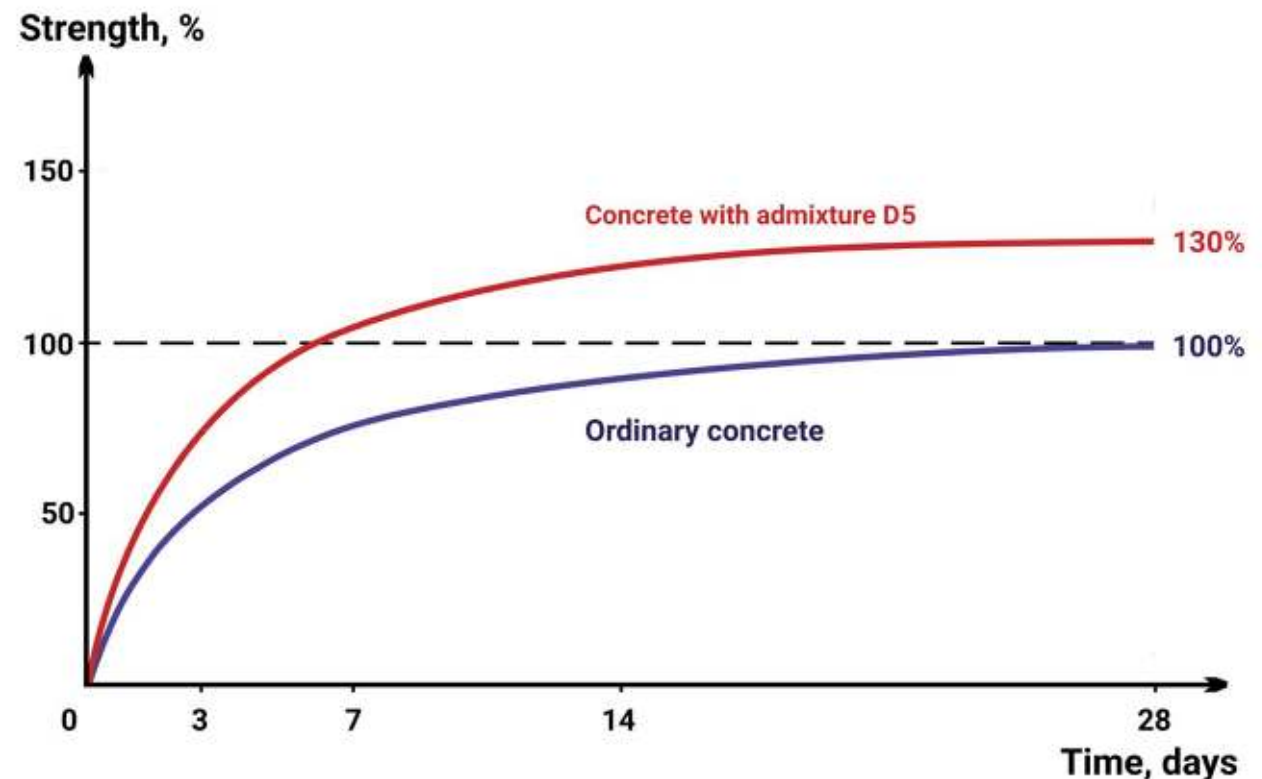
Leading Researcher
Head of the Laboratory

V.N. Khokhlov
A.A. Kuprin

Наименование лаборатории	ИД "ИВ-Стройинжиниринг" Инжендерно-испытательный центр (закрытого типа)		
в составе	ООО "ВНИСТРОМ-НВ" 143050, г. Красногорск, Московская обл., ул. К. Маркса, 117 наименование юридического лица		
Аттестат аккредитации №	РОСС RU.000.1310407		
ПРОТОКОЛ ИСПЫТАНИЙ № 157И			
		от 19 октября 2016 года	
Наименование продукции	Композитная полиэфирная цементная добавка «Д-5»		
Заказчик	ООО «ПНП «ТОКАР» 342302, г. Владикавказ, ул. Петуриной, д. 19 «а»		
Составляющие испытательного образца	Объемная проба добавки «Д-5» в пористообъемном виде – 1 кг (длина изготовления: 20.09.2016 г., серия № 35).		
Составляющие испытательного образца	Состав бетонной смеси: Цем I 42,5 – 353 кг/м³; песок Мкр 2 – 850 кг/м³; щебень гранитный фракция 5 до 20 мм – 1100 кг/м³; добавка Д-5 – 10,5 кг/м³ (3 % от массы цемента), кол-во – 173 кг/м³, В/Ц – 0,486. Наименование бетонной смеси ПЗ (ОК – 14 см). Бетонную смесь укладывали в образки, уплотняли уплотнителем и выдерживали 28 суток в нормальных условиях (W > 90 %, t = 21 °C).		
Регистрационные данные испытательного центра (лаборатория)	16.145(ИД)		
Цель испытаний	Определение показателей «водостойкости» бетона при дозировке добавки 3 % от массы цемента.		
Методика испытаний	ГОСТ 12730.5-84 «Цемент. Методы определения водонепроницаемости». 1.2 «Определение водонепроницаемости по методу «песку». Бетонные образцы в образки или кубы нект вы укладывают в форму высотой 150 мм, с диаметром 150 мм. Установка имеет 6 ярусов для увеличения образцов в обеспечении возможности, когда вода к нижней ступени непосредственно образки при регулярном израсходо-е до 4 МПа (с 60 мм) диаметра. Вертикаль толщина пористости образков, диаметр 140 мм диаметр для образков. Давление воды повышали ступенчато по 0,2 МПа (2 атм) с выдержкой на каждой ступени 24 часа.		
Условия испытаний	t = 21 °C, W = 81 %		
Дата поступления	27.07.2016 г.		
Дата испытаний	27.07.2016 г. – 16.09.2016 г.		
РЕЗУЛЬТАТЫ ИСПЫТАНИЙ:	1. Испытанные образцы с добавкой Д-5 в количестве 3 % от массы цемента выдержали давление воды 3,5 МПа (35 атм) без признаков фильтрации воды. Высота поднятой воды бетона в образках при их расслоении от 30 см 32 мм.		
Исполнитель: инженер научный сотрудник	В.П. Хохлов		
Руководитель испытательной лаборатории:	А.А. Кутянов		

The effect of D5 on strength and rate of strength gain

- Accelerates strength gain of concrete almost by 4 times: **100% normative strength of concrete is achieved within 5-7 days of hardening under normal conditions**
- Increases the strength of concrete at the age of 28 days by **30%** as compared to concrete without admixtures (in mixes with equal slump)

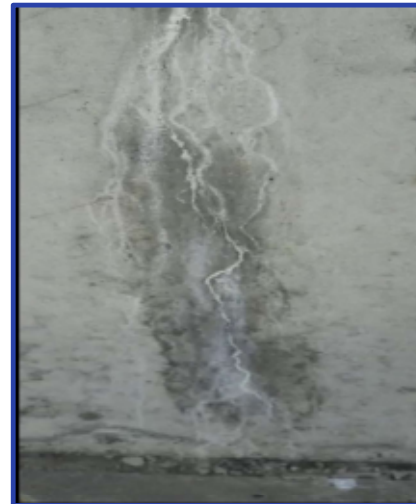


Self-sealing capacity of through cracks up to 0.5 mm wide

Self-sealing of through cracks up to 0.5 mm wide, which may appear in constructions due to low-quality concreting, the differential settlement of buildings or because of the dynamic impact (including seismic load), occurs in concretes with the admixture D5. Water flow through cracks is the essential condition for self-sealing.

«Cold» joints do not leak!

Due to the self-sealing effect, **«cold» joints in concretes with the admixture D5 do not leak even with long breaks in concreting** (up to 15 days and more). Simply clean the hardened concrete surface from cement crust, expose the aggregate structure of the concrete, wash off all dust and thoroughly moisten the surface. Before pouring fresh concrete, the joint should be moistened again and any dust removed. Under these conditions, the fresh concrete bonds with the hardened concrete at the molecular level.



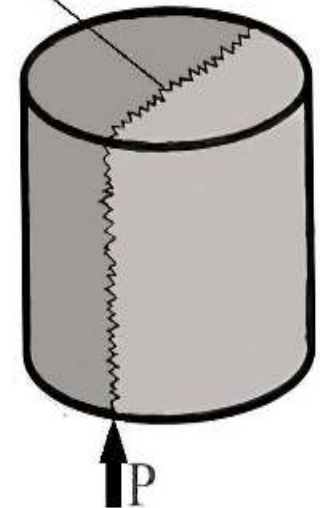
Cold joints in concrete with **D5** exhibit no leakage at pressures of up to 12 bar

Tests were conducted at the Belarusian National Technical University (Test report #302, dated 18.05.2022)

- To test the waterproofing of cold joints in concrete with the admixture D5, the cylindrical samples (150 mm in diameter and 150 mm high) were cast.
- The samples were made in 2 stages: first, a cylinder half (semi-cylinder) was cast (See Fig.). After 24 hours, the hardened half of the cylinder was removed from the mold, the vertical surface of the cold joint was cleaned from cement crust and moistened.
- The cylinder half was then placed back into the mold and cast with ready-mixed concrete with D5, creating a concrete cylinder with a vertical cold joint.
- Six cylindrical samples with cold joints were cast. They were cured in a standard curing room for 28 days, then subjected to a water permeability test (wet spot method) in accordance with GOST 12730.5-2018.
- The water pressure was increased in increments of 0.2 MPa every 1-2 minutes, with each stage maintained for 24 hours.
- **Testing demonstrated that cold joints in samples of concrete with D5 remained watertight under pressures of up to 12 bar (1.2 MPa).**

Cold joint

Холодный шов



Additional properties

- The admixture D5 is non-aggressive toward reinforcing steel and does not cause its corrosion. It enhances passivating characteristics of concrete toward reinforcing steel and is recommended for use in critical reinforced concrete structures.
- The admixture D5 **increases adhesion between concrete and reinforcement by 36%.**
- The admixture D5 **increases adhesion to aged concrete by 1.9 times or 90%** compared to concrete without admixtures.

The NIIZhb Report

НАУЧНО-ИССЛЕДОВАТЕЛЬСКИЙ, ПРОЕКТНО-КОНСТРУКТОРСКИЙ И
ТЕХНОЛОГИЧЕСКИЙ ИНСТИТУТ БЕТОНА И ЖЕЛЕЗОБЕТОНА
ФИЛИАЛ ФГУП «НИИ «СТРОИТЕЛЬСТВО»
(«НИИЖБ»)

УТВЕРЖДАЮ
Директор НИИЖБ
Семченко А.С.
2006г.

Заключение
по результатам испытаний новой универсальной добавки
к бетону серии «Д» на коррозию арматуры в бетоне,
сцепление арматуры с бетоном и адгезию к старому бетону
(х/д № 327/13-109-06/ЖБ)

Зав. лабораторией коррозии и долго-
вечности бетонных и железобетонных
конструкций, д.т.н., проф.

Научные сотрудники

Степанова В.Ф.
Звягина Т.Л.
Харитонова Л.П.

Test results on chloride-ion permeability for concrete with the admixture D5

➤ Concrete with the admixture D5 has very low chloride-ion permeability

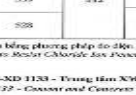
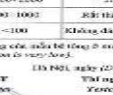
438 < 1000 at a dosage of 3%
512 < 1000 at a dosage of 2%

VIETNAM INSTITUTE FOR BUILDING MATERIALS (VIBM)

LAS-XD 1133 LABORATORY Cement and Concrete Center

Nguyen Van Dan

 BỘ XÂY DỰNG VIỆN VẬT LIỆU XÂY DỰNG PHÒNG THÍ NGHIỆM LAS-XD 1133 (SỐ QUÂN 17065.2017) - TRUNG TÂM XI MĂNG & BÊ TÔNG VIETNAM INSTITUTE FOR BUILDING MATERIALS (VIBM) LAS-XD 1133 LABORATORY - CEMENT AND CONCRETE CENTER Địa chỉ (Address): 255 Đường Nguyễn Trãi P. Thanh Xuân Trung Q. Thanh Xuân, TP. Hà Nội Điện thoại (Tel): 024.38589115; Fax: 024.38581112; Email: chieudat@vibm.vn; Web: www.vibm.vn						
PHIẾU KẾT QUẢ THỬ NGHIỆM TEST REPORT Số phiếu (Doc. No.): VN-XD-XMHT Số phiếu (Doc. No.): VN-XD-XMHT-HĐK-368-Tiến hành thử 42						
1. Đơn vị yêu cầu (Client):	CÔNG TY CỔ PHẦN TẬP ĐOÀN K&S					
2. Địa chỉ (Address):	SỐ 62 NGUYỄN HUY TƯỜNG, QUẬN THANH XUÂN, TP. HÀ NỘI					
3. Dự án (Project):	PHỤ GIA D5 SỬ DỤNG CHO BÊ TÔNG NƯỚC BIỂN					
4. Loại mẫu (Kind of sample):	Bê tông mác 102-200 (mm)					
5. Phương pháp thử (Test method):	TCVN 9337:2012					
6. Ngày thử mẫu (Sampling date):	29/10/2020					
7. Ngày thí nghiệm (Date of testing):	04/12/2020					
KẾT QUẢ ĐỘ CHỐNG THẨM ION CLO BẰNG PHƯƠNG PHÁP ĐO ĐIỆN LƯỢNG RESULT FOR CONCRETE'S ABILITY TO RESIST CHLORIDE ION PENETRATION						
Số (No.)	Ký hiệu mẫu (Mark of sample)	Đơn vị (Unit)	Kết quả (Result)		Bảng giá theo TCVN 9337:2012 (Classification according to TCVN 9337:2012)	
			Độ thấm qua mẫu (Charge Passed)	Trung bình (Average)	Độ thấm qua mẫu (Charge Passed)	Mức độ thẩm thấu (Permeability Class)
M1	Mẫu thử phụ gia + nước biển	Cường (Cubic)	438	438	>4000	Cao (High)
500-9000			Trung bình (Medium)			
100-1000			Rất thấp (Very Low)			
M2	412	412	<100	Không đáng kể (Negligible)		
M3	447	447	<100	Không đáng kể (Negligible)		
*Nhận xét (Remarks): Kết quả độ chống thấm ion clo bằng phương pháp đo điện lượng của mẫu bê tông ở mức độ thấp <i>(Result for Concrete's ability to Resist Chloride Ion Penetration is very low).</i>						
Viện Vật liệu xây dựng VIBM		PTN LAS-XD 1133 - Trung tâm XM&BT LAS-XD 1133 - Cement and Concrete Center		Hà Nội, ngày (Date): 04/12/2020		
				 		
Chú thích (Note): Mẫu bê tông thử nghiệm đã được VIBM kiểm tra đúng quy trình theo tiêu chuẩn TCVN 9337:2012. Tất cả các kết quả thử nghiệm đều được ghi chép vào sổ biên bản thử nghiệm và gửi kèm theo báo cáo. Mọi thắc mắc về kết quả thử nghiệm xin vui lòng liên hệ với phòng thí nghiệm để được giải đáp. Không được sao chép thông tin này để sử dụng cho mục đích khác ngoài mục đích được cấp phép.						

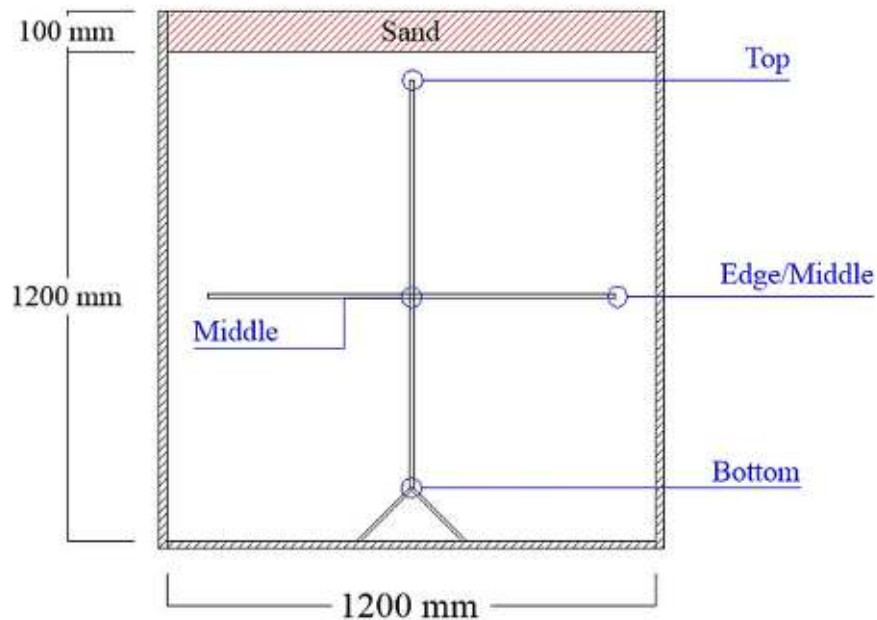
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M2	512	512	100-1000	Rất thấp (Very Low)		
M3	528	528	<100	Không đáng kể (Negligible)		
*Nhận xét (Remarks): Kết quả độ chống thấm ion clo bằng phương pháp đo điện lượng của mẫu bê tông ở mức độ thấp <i>(Result for Concrete's ability to Resist Chloride Ion Penetration is very low).</i>						
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The admixture **D5** does not raise the temperature of concrete in mass concrete structures

Comparative tests for concrete heat release were conducted in Australia in accordance with British Standards

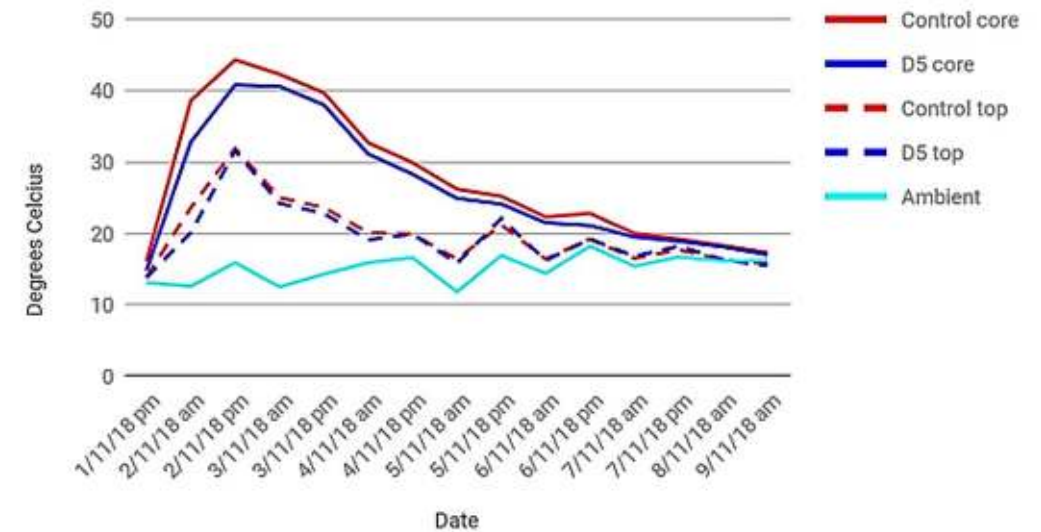
The test results were issued by NEOCRETE Co., Ltd, our official distributor in New Zealand and Australia

Temperature sensors layout



Temperature profile of concrete

Heat generated (cubes)



Plasticizing and water-retaining properties, as well as improved pumpability of concrete mixes

- The admixture D5 either increases flowability of concrete mixes from P1 to P5 or **reduces water consumption by 20-30%** in mixes with equal slump
- The admixture D5 provides water retention, prevents segregation and improves pumpability

Construction of a waterproof concrete foundation slab using the admixture D5 near Nelson, New Zealand



Frost-resistance and air entrainment



Enhances frost-resistance by at least two grades, enabling concrete to achieve a rating **exceeding F400**



Increases air entrainment of concrete by up to 2-3% for CEM I cements and by up to **4-6%** for CEM II cements containing slag

Eco-friendliness

- The admixture D5 contains no components that negatively affect cement stone and reinforced structures, **nor does it form toxic compounds in air or water**
- The admixture D5 is approved for production of concrete and reinforced concrete structures contacting drinking water, as well as in water treatment systems
- The admixture D5 is used in construction of swimming pools and fish-breeding tanks
- The admixture D5 has **State Registration Certificate** No. RU.15.01. 09.013. E000008.08.15
- The admixture D5 has the **Ecological Certificate of Conformity** No. POCC RU.31545.04ИЗЕО.РЭС-134 from May 25, 2021



Test results of the admixture **D5** in Australia and New Zealand in accordance with British Standards

1. Tests on concrete strength at the age of 1, 3, 7, 28, 56 and 416 days

Sample	1 day	3 days	7 days	28 days	56 days	416 days
Control concrete (without D5)	9 (MPa)	22,5 (MPa)	31,5 (MPa)	43,5 (MPa)	48,5 (MPa)	57 (MPa)
2% D5	15,5 (MPa)	27,5 (MPa)	40 (MPa)	54 (MPa)	56 (MPa)	67 (MPa)
3% D5	22 (MPa)	37 (MPa)	47,5 (MPa)	58,5 (MPa)	62 (MPa)	78,5 (MPa)



2. Water permeability testing of concrete with D5 under 5 bar of pressure over a 3-day period

- The maximum water penetration depth was recorded at **5 mm**



D5 Admixture: Recommendations for Use

- The recommended dosage of the admixture D5 is **2-3% of the total cement weight**, subject to the required concrete properties and results of preliminary testing with local cements and aggregates;
- D5 is introduced into a concrete mixer **in a dry form** together with sand or cement;
- Mixing time is **60-120 seconds**;
- **If there are interruptions in concreting**, the surface of the previously poured concrete (with the admixture D5) should be cleaned of cement crust immediately after hardening (6-8 hours after pouring) at the location of the future «cold» joint. It is necessary to expose the aggregate, wash away dust and debris, and thoroughly moisten the concrete at the joint area;
- **After breaks in concreting** (lasting days to weeks), the «cold» joint surface must be re-cleaned to remove dust and debris, immediately before pouring the concrete mix with the admixture D5;
- Under these conditions, D5 ensures high adhesion and molecular-level fusion between hardened and fresh concrete, providing superior waterproofing for cold technological joints (both vertical and horizontal).



**Vertical and horizontal
«cold» joints do not leak**

Packing

- **15 kg valve paper bags**, palletized, wrapped in stretch film and secured with strapping
- **1000 kg big-bags**, palletized and secured with strapping



Shelf life

- Shelf life is at least **36 months** in original packaging

We guarantee the high quality of our products



An ISO-9001-2015 Quality Management System has been implemented in the company



We have our own test laboratory, as well as R&D and quality control departments



Patented technologies for the production of concrete admixtures



We ensure product quality control at every stage of production



Highly skilled professionals

High quality is certified by the certificates:

- Russian Certificate of Conformity
- New Zealand Certificate of Conformity
- ISO Compliance Certificate
- Certificate of State Registration
- Radiation Safety Certificate
- Environmental Certificate



Our largest-scale project with D5 is the construction of a 14.2 km derivation tunnel at the Zaramag Hydroelectric Power Station in North Ossetia – Alania. More than 140,000 m³ of hydraulic concrete using D5 were poured.



The admixture D5 can be applied in various construction projects



Swimming pools



Drinking water tanks

The admixture D5 can be applied in various construction projects



Oil-storage tanks



Treatment facilities

The admixture D5 can be applied in various construction projects



Underground parking



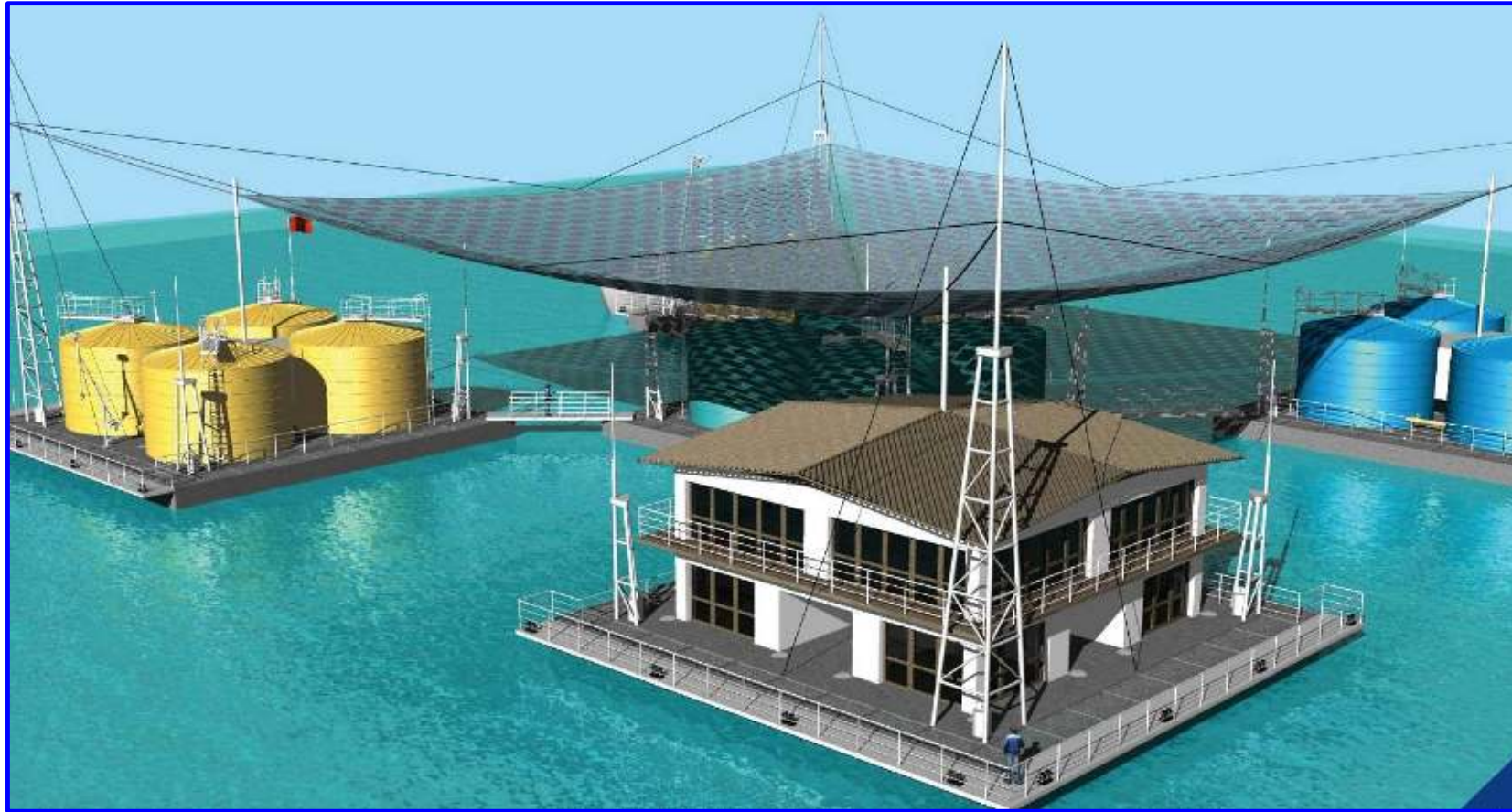
Underground and underwater tunnels

The admixture D5 can be applied in various construction projects



Fish-breeding ponds

The admixture D5 can be applied in various construction projects



Floating platforms for various purposes

The admixture D5 can be applied in various construction projects



Floating platforms for various purposes

The admixture D5 can be applied in various construction projects



Floating platforms for various purposes

WHAT IS SULFATE CORROSION OF CONCRETE?

- Let's examine the process of sulfate attack on concrete in aggressive environments:

1. It is known that sulfate ions in water penetrate into the concrete matrix through microcracks and react with free lime to form gypsum



2. The gypsum reacts with the tricalcium aluminate of cement stone ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$) in the presence of water



3. The formation of ettringite crystals causes significant volumetric expansion, resulting in internal tensile stresses in concrete followed by splitting of cement stone

EXAMPLES OF SULFATE ATTACKS ON CONCRETE



HOW TO PREVENT SULFATE CORROSION IN CONCRETE?

1. It is necessary to prevent penetration of water with dissolved sulfate-ions into the concrete body by maximizing the water resistance of concrete
2. It is necessary to bind free lime by introducing an active pozzolanic additive so that during **pozzolanic reaction** the free lime is bound by amorphous silica to form a strong, poorly soluble compound – **calcium silicate**



The admixture D5 successfully meet both of these conditions: it gives concretes the highest possible water resistance and, being an active pozzolanic additive, binds free lime in the cement stone. Therefore, **concretes with D5 are sulfate-resistant even when using ordinary Portland cement.**

DEPENDENCE OF SULFATE RESISTANCE OF CONCRETE ON ITS WATER RESISTANCE

- Based on numerous studies, the **Research Institute of Reinforced Concrete (NIIZhB)** has identified a direct correlation between the sulfate resistance of concrete on its water resistance. It turns out that the same aggressive environment has a different aggressive impact on concretes with different water resistance. This correlation is illustrated in GOST 31384-2017.

The degree of aggressive impact from soils, liquids, and biological environments

Diagram B.1 – The degree of aggressive impact from sulfates in soils on concrete grades based on water-resistance properties W4-W20

Cement		Soil aggressiveness criteria, mg/kg, for concrete grade based on water resistance					The degree of aggressive soil impact on concrete
Cement groups classified based on sulfate-resistance	Cement type	W4	W6	W8	W10 – W14	W16 – W20	
I	Portland cements according to GOST 10178, ГОСТ 31108, not included in group II	over 500 to 1000	over 1000 to 1500	over 1500 to 2000	over 2000 to 3000	over 3000 to 4000	Mild aggressiveness
		over 1000 to 1500	over 1500 to 2000	over 2000 to 3000	over 3000 to 4000	over 4000 to 5000	Moderate aggressiveness
		over 1500	over 2000	over 3000	over 4000	over 5000	Severe aggressiveness
II	Portland cements according to GOST 10178, ГОСТ 31108 with clinker content of C ₃ S of no more than 65%, of C ₃ A of no more than 22% and slag Portland cements	over 3000 to 4000	over 4000 to 5000	over 5000 to 8000	over 8000 to 10000	over 10000 to 12000	Mild aggressiveness
		over 4000 to 5000	over 5000 to 8000	over 8000 to 10000	over 1000 to 12000	over 12000 to 15000	Moderate aggressiveness
		over 5000	over 8000	over 10000	over 12000	over 15000	Severe aggressiveness
III	Sulfate-resistant concretes according to GOST 22266	over 6000 to 8000	over 8000 to 10000	over 10000 to 12000	over 12000 to 15000	over 15000 to 20000	Mild aggressiveness
		over 8000 to 10000	over 10000 to 12000	over 12000 to 15000	over 15000 to 20000	over 20000 to 24000	Moderate aggressiveness
		over 10000	over 12000	over 15000	over 20000	over 24000	Severe aggressiveness

WHY CONCRETES WITH **D5** ON ORDINARY PORTLAND CEMENT ARE SULFATE-RESISTANT?

- **Clause 7.4.1** of GOST 31384 explicitly states that Portland cements and Portland cements with multifunctional additives should be used primarily when producing concrete for aggressive environments. Only if these measures prove insufficient, should sulfate-resistant cements be used.
- **Clause 7.4.3** of GOST 31384 states that to increase the durability of concrete in reinforced concrete structures operating in aggressive environments, mineral admixtures should be used that reduce permeability, increase frost resistance, and do not have a corrosive effect on concrete and reinforcement. Mineral admixtures should be used in combination with plasticizing and water-reducing admixtures. The total amount of additives should not exceed 5% of the cement mass.
- The admixture D5 additive is a mineral additive that reduces permeability, increases frost resistance, has a plasticizing and water-reducing effect, and does not corrode concrete or reinforcement (NIIZhB report). The recommended dosage is 3% of the cement weight.

As we can see, in full compliance with GOST 31384-2017 «Protection of concrete and reinforced concrete structures from corrosion», CONCRETES WITH THE ADMIXTURE D5 PRODUCED USING TYPE I AND II PORTLAND CEMENTS ARE CONSIDERED SULFATE-RESISTANT OFFERING RELIABLE AND DURABLE PRIMARY PROTECTION AGAINST AGGRESSIVE ENVIRONMENTS THROUGHOUT THE ENTIRE SERVICE LIFE OF STRUCTURES!

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